

PROBLEM SET 2—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Instructions: Due on 09/10/19 at the beginning of the lecture. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission. Don't forget to attach the cover sheet.

1. Consider the model we saw in class about credit rationing. For completeness sake, I write here again the main description. Consider a population of risk-neutral borrowers who each own a project that requires an initial outlay of $I = 1$ and yields a random return X , where $X \in \{0, R\}$. Let $p \in [0, 1]$ denote the probability that $X = R$. Borrowers have no wealth and must obtain investment funds from a bank. A borrower can be of two different types $i = s, r$. The borrower of type i has project with return characteristics (p_i, R_i) . Assume the following.

A1' $p_r R_r < 1 < p_s R_s$.

A2 $p_s > p_r$ and $R_s < R_r$.

As you can see, assumption A1' is different from the one we covered in class. Assume that there is excess demand for funds for the bank, i.e. $\alpha < 1$ and that available funds are sufficient to avoid crowding out either type of borrower completely, i.e. $\max\{\beta, 1 - \beta\} < \alpha$.

- (a) Compute the optimal single fixed repayment scheme D for the bank. Is there any type of borrower that is excluded for sure?
- (b) Now consider that the bank can offer a type-dependent contract that specifies the probability of financing, x_i and the repayment amount D_i if the project is successful. That is, the bank can offer a plan (x_i, D_i) for $i = s, r$. Compute the optimal contracts for the bank under assumptions A1' and A2. Show that in contrast with what we saw in class, now credit rationing reappears under two possible forms:
 - If $R_s[\beta p_s + (1 - \beta)p_r] \geq 1$ then both types of project are funded and there is cross-subsidization between types. Conclude that this happens because the banks cannot turn away risky borrowers without restricting access to credit for the safe borrowers.
 - If $R_s[\beta p_s + (1 - \beta)p_r] < 1$ then neither project is funded, which is a severe form of credit rationing, i.e. *total financial collapse*.

Solution

- (a) Let D be a single-fixed repayment scheme. As seen in the lecture notes, type- i borrowers apply for funds if and only if $D \leq R_i$. If $D > R_s$ only the risky borrowers will consider applying. The maximum then that the bank can charge is $D = R_r$. In this case, the bank is only contracting with risky borrowers and obtains $(1 - \beta)(p_r R_r - 1)$. In contrast with the case analyzed in class, assumption A1' implies that this term is negative. Thus, contracting with risky borrowers only is not optimal. This implies that $D \leq R_s$, i.e. the bank should contract with both types, assuming of course that this is profitable. If this is the case, the bank should charge the highest possible repayment that the safe borrowers can afford, that is $D = R_s$. In this case, the profits obtained by the bank are given by

$$\alpha(\beta(p_s R_s - 1) + (1 - \beta)(p_r R_s - 1)).$$

In order for the bank to be willing to contract with both borrowers, the expression above must be positive. Thus, profits are positive if and only if $R_s(\beta p_s + (1 - \beta)p_r) \geq 1$ and in this case the bank finds optimal to contract with both types. On the other hand, if $R_s(\beta p_s + (1 - \beta)p_r) < 1$ the bank abstains from lending to any of the borrowers. Note that in both situations, there is credit rationing.

- (b) We need to characterize the second-best contract for the bank. A contract is defined by (x_i, D_i) where

x_i is the probability of obtaining financing and D_i is the amount to be repaid. The bank's problem is:

$$\max_{(x_i, D_i)_{i=s,r}} \{ \beta x_s (p_s D_s - 1) + (1 - \beta) x_r (p_r D_r - 1) \} \quad (1)$$

$$\text{subject to } 0 \leq x_i \leq 1, \quad \forall i = s, r \quad (2)$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (3)$$

$$x_i p_i (R_i - D_i) \geq 0 \quad \forall i = s, r \quad (4)$$

$$x_i p_i (R_i - D_i) \geq x_j p_i (R_i - D_j) \quad \forall i, j = s, r \quad (5)$$

Because of assumption A2, the risky type still has the highest willingness to repay, so it is still considered the "high type". Therefore, the previous problem is equivalent to the reduced problem:

$$\max_{(x_i, D_i)_{i=s,r}} \{ \beta x_s (p_s D_s - 1) + (1 - \beta) x_r (p_r D_r - 1) \} \quad (6)$$

$$\text{subject to } x_s p_s (R_s - D_s) \geq 0 \quad (7)$$

$$x_r p_r (R_r - D_r) \geq x_s p_r (R_r - D_s) \quad (8)$$

$$x_r \geq x_s \quad (9)$$

$$0 \leq x_i \leq 1 \quad i = s, r \quad (10)$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (11)$$

Furthermore, we know that the participation constraint for the safe type and the incentive constraint for the risky type will be binding. Solving for D_s and D_r using these binding constraints and plugging the resultant expressions into the objective yields the following problem:

$$\max_{x_s, x_r} x_s (\beta (p_s R_s - 1) - (1 - \beta) p_r (R_r - R_s)) + x_r ((1 - \beta) (p_r R_r - 1)) \quad (12)$$

$$\text{subject to } 0 \leq x_s \leq x_r \leq 1 \quad (13)$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (14)$$

Notice that in the objective (12), the term attached to x_r is strictly negative. Thus the bank would like to shutdown any interaction with the risky types. However, if $x_r = 0$ then it would force x_s to be zero also because of the monotonicity constraint. Since it might still be profitable to contract with the safe type, then $x_s = 0$ might not always be optimal. This implies that the monotonicity constraint will be binding. To see this, suppose towards a contradiction that $\hat{x}_s$ and $\hat{x}_r$ are optimal solutions such that $\hat{x}_r = \hat{x}_s + \epsilon$ where $\epsilon > 0$. The expected profit then is

$$\hat{x}_s [R_s (\beta p_s + (1 - \beta) p_r) - 1] + \epsilon (1 - \beta) (p_r R_r - 1).$$

However, by taking instead $\tilde{x}_r = \hat{x}_s$ we have that the profit from the contract $(\hat{x}_s, \tilde{x}_r)$ is such that

$$\hat{x}_s [R_s (\beta p_s + (1 - \beta) p_r) - 1] + \epsilon (1 - \beta) (p_r R_r - 1) < \hat{x}_s [R_s (\beta p_s + (1 - \beta) p_r) - 1]$$

which contradicts the optimality of $\hat{x}_r$. Thus, the monotonicity constraint must be binding.

Let $x = x_r = x_s$. Then the problem reduces to

$$\max_{x \in [0, \alpha]} x [R_s (\beta p_s + (1 - \beta) p_r) - 1] \quad (15)$$

From the previous problem it is easy to see that we have the following optimal solution

$$x^* = \begin{cases} \alpha & \text{if } R_s (\beta p_s + (1 - \beta) p_r) \geq 1, \\ 0 & \text{if } R_s (\beta p_s + (1 - \beta) p_r) < 1. \end{cases}$$

With this solution, it is easy to find the optimal repayment scheme from the binding incentive and participation constraints: $D_r = D_s = D^* = R_s$. Thus, the optimal contract is a *pooling contract* and it

is the same as the one found in the previous item. Furthermore, the solution implies that if the bank decides to actually fund any of the consumers, it is cross-subsidizing the risky types with the profits obtained from the safe ones. If this is the case, there is still credit rationing because the probability of funding any type of consumer is α . When the bank decides not to contract with any type, there is a total financial collapse, since no consumer gets credit.

2. (J. Tirole) Assume that there are two types of consumers for a firm's product, θ_H and θ_L . The proportion of type θ_L consumers is λ . A type θ 's utility when consuming amount x of the good and paying a total T for it is $u(x, T) = \theta v(x) - T$, where

$$v(x) = \frac{1 - (1 - x)^2}{2}.$$

The firm is the sole producer of this good, and its cost of production per unit is $c > 0$.

- Consider a non-discriminating monopolist. Derive his optimal pricing policy, i.e. the cost per unit that the monopolist will charge. Show that he serves both classes of consumers if either θ_L or λ is "large enough." *Hint: Derive the demand curve for each type as a function of the price charged by the monopolist and then compute the "aggregate demand". Use this demand to solve for the optimal price of the monopolist.*
- Consider a monopolist who can distinguish the two types (by some characteristic) but can only charge a simple price p_i for each type θ_i . Characterize the optimal prices. Show that at the optimal prices, the monopolist is charging a marginal price $p = c$ per unit plus a personalized fixed fee equal to the consumer's *net surplus*.
- Suppose that the monopolist cannot distinguish the types. Suppose that the monopolist can charge a single *two-part tariff*. A two part tariff is an *affine pricing schedule*: $T(q) = A + pq$, where A is a fixed fee and p is a marginal price per unit. Derive the optimal two part tariff under the assumption that the monopolist serves both types.
- Compute the fully optimal nonlinear tariff (second best optimum for the monopolist's screening problem). How do the quantities purchased by the two types compare with the levels obtained in (a)–(c). How do the monopolist's profits from each case in (a)–(d) compare?

Solution

It will be useful to calculate some expressions that will be used repeatedly. For each type, the first order condition in the utility maximization problem of the consumer when he faces a *single-posted price* yields the individual demands:

$$\theta_i(1 - x_i) = p_i \Rightarrow x_i(p) = \max \left\{ 1 - \frac{p}{\theta_i}, 0 \right\} \quad \text{for } i = L, H. \quad (16)$$

Assume that $\theta_H > \theta_L > c > 0$ in order to simplify the analysis by eliminating the cases in which no consumer would like to buy at the lowest price the monopolist can produce without incurring in losses.

In order to simplify calculations, let θ denote the *harmonic mean* of θ_L and θ_H :

$$\frac{1}{\theta} \equiv \frac{\lambda}{\theta_L} + \frac{1 - \lambda}{\theta_H}. \quad (17)$$

Then the aggregate demand at any price p can be written as

$$X(p) = \lambda x_L(p) + (1 - \lambda)x_H(p) = 1 - \frac{p}{\theta}. \quad (18)$$

Net consumer surplus at a price p is given by

$$S_i(p) = \theta_i v(x_i(p)) - p x_i(p)$$

so that for the specific functional form that we assumed for the utility the previous expression yields

$$S_i(p) = \theta_i \left(\frac{1 - (1 - x_i(p))^2}{2} \right) - p x_i(p) = \frac{(\theta_i - p)^2}{2\theta_i}. \quad (19)$$

Note that $S_i \geq 0$ and that surplus is always higher for θ_H types.

- (a) The monopolist that can only charge a single price p would solve the following problem

$$\Pi_1 = \max_{p \geq 0} (p - c)X(p) \quad (20)$$

Since $X(p) \leq 0$ for any $p \geq \theta$ and the objective is strictly concave in p , the first order condition is necessary and sufficient to identify a solution to the monopolist's problem, which yields the monopoly price

$$p_1 = \frac{\theta - c}{2}$$

and the monopoly profits

$$\Pi_1 = \frac{(\theta - c)^2}{4\theta}.$$

Note that this solution assumes that the monopolist chooses to serve the two types of consumers. The monopolist could potentially choose a price high enough such type θ_L consumers choose not to buy. Such a strategy is optimal if the monopoly price relative to this category, which is $\frac{c + \theta_H}{2}$, exceeds θ_L and if the profits from this strategy are larger than the monopoly profits of serving the two types.

Serving only the high type at monopoly price $(c + \theta_H)/2$ yields

$$\Pi'_1 = (1 - \lambda) \frac{(\theta_H - c)^2}{4\theta_H}.$$

It can be shown that^a $\frac{\partial(\Pi_1 - \Pi'_1)}{\partial \lambda} > 0$. Furthermore, it is easy to see that $\lim_{\lambda \rightarrow 0} (\Pi_1 - \Pi'_1) < 0$ and that $\lim_{\lambda \rightarrow 1} (\Pi_1 - \Pi'_1) > 0$ and thus, by monotonicity and the intermediate value theorem, $(\Pi_1 - \Pi'_1)$ is equal to zero at just one point λ^* . Thus, for $\lambda \geq \lambda^*$, i.e. if λ is "high enough", serving the two types of consumers is more profitable than only serving the high types. Alternatively, if $\theta_L > (c + \theta_H)/2$, monopoly pricing for the type θ_H consumers yields consumption for the type θ_L consumers.

- (b) If the monopolist can distinguish between the two types then we are calculating the first-best outcome. Thus, the pair of problems of the perfectly discriminating monopolist are given by

$$\max_{x_i, T_i} T_i - c x_i \quad \text{subject to } \theta_i v(x_i) - T_i \geq 0 \text{ for } i = L, H.$$

As usual, the participation constraint must be binding and we can simplify the pair of problems to the following ones:

$$\max_{x_i, T_i} \theta_i v(x_i) - c x_i \quad \text{subject to } \theta_i v(x_i) - T_i \geq 0 \text{ for } i = L, H.$$

Notice that the monopolist is maximizing social surplus for each type. Due to the specific form for v that we assumed, the objective is strictly concave and continuous and therefore, the first order conditions are necessary and sufficient. From them we get

$$\theta_i (1 - x_i^f) = c \Rightarrow x_i^f = 1 - \frac{c}{\theta_i}, \quad i = L, H.$$

At this solution, we have that $T_i^f = \theta_i v(x_i^f) = \frac{(\theta_i - c)(\theta_i + c)}{2\theta_i}$. Notice that this transfer is equivalent to the monopolist charging a marginal price of $p = c$ per unit and also charging a fixed fee equal to the net surplus of each consumer $S_i(c) = \frac{(\theta_i - c)^2}{2\theta_i}$. Notice that

$$S_i(c) + c x_i^f = \frac{(\theta_i - c)^2}{2\theta_i} + c x_i(p) = \frac{(\theta_i - c)^2}{2\theta_i} + \frac{\theta_i c - c^2}{\theta_i} = \frac{\theta_i^2 - c^2}{2\theta_i} = \frac{(\theta_i - c)(\theta_i + c)}{2\theta_i} = T_i^f.$$

Profits from the first best are

$$\Pi_2 = \lambda(T_L^f - cx_L^f) + (1 - \lambda)(T_H^f - cx_H^f) = \frac{\lambda(\theta_L - c)^2}{2\theta_L} + \frac{(1 - \lambda)(\theta_H - c)^2}{2\theta_H}.$$

- (c) Now we look for the optimal single two-part tariff. Again, for simplicity, we will assume that the monopolist serves the two types of consumers, i.e. that λ and θ_L are sufficiently large, as we saw before.

Suppose that the monopolist charges a marginal price of p . The highest fixed fee that is consistent with type θ_L consumers buying the good is $A = S_L(p)$. Type θ_H consumers then buy because as mentioned before $S_H(p) > S_L(p) = A$. Thus, the monopolist's problem is

$$\max_{p \in [0, \infty)} S_L(p) + (p - c)X(p).$$

The monopolist will do at least as well under the optimal single two-part tariff as under a fully linear tariff (single monopoly price). To the variable profit $(p - c)X(p)$ the monopolist is adding the fixed fee $S_L(p)$ that is paid by all types of consumers. Alternatively, notice that a linear tariff is a special case of a two-part tariff.

The first order condition yields

$$\frac{p - \theta_L}{\theta_L} - \frac{p}{\theta} + \frac{c}{\theta} + 1 - \frac{p}{\theta} = 0.$$

Solving the previous equation yields $p_3 = \frac{c}{2 - \frac{\theta}{\theta_1}}$. The monopolist profit, after a lot of tedious algebra is

$$\Pi_3 = \frac{1}{4} \left(c^2 \left(\frac{1}{\theta} + \frac{1}{2\theta_L - \theta} \right) - 4c + 2\theta_L \right).$$

- (d) Finally, for the optimal non-linear tariff, we follow the standard procedure. From the general problem analyzed in class, we know that the monopolist will solve

$$\max_{(x_i, T_i)_{i=L, H}} \lambda(T_L - cx_L) + (1 - \lambda)(T_H - cx_H) \quad (21)$$

$$\text{subject to } \theta_H v(x_H) - T_H \geq \theta_H v(x_L) - T_L \quad (22)$$

$$\theta_L v(x_L) - T_L \geq \theta_L v(x_H) - T_H \quad (23)$$

$$\theta_H v(x_H) - T_H \geq 0 \quad (24)$$

$$\theta_L v(x_L) - T_L \geq 0 \quad (25)$$

$$x_L \geq 0, x_H \geq 0 \quad (26)$$

This problem is equivalent to the reduced problem in which we drop the incentive constraint of the low type, the participation constraint of the high type and we impose a monotonicity constraint. Furthermore, the incentive constraint of the high type and the participation constraint of the low type need to be binding. Using these facts and substituting the binding constraints into the objective and rewriting it we can solve the following problem:

$$\max_{x_L \geq 0, x_H \geq 0} \theta_L v(x_L) - cx_L + \frac{1 - \lambda}{\lambda} (\theta_H v(x_H) - cx_H) - \frac{1 - \lambda}{\lambda} (\theta_H v(x_L) - \theta_L v(x_L))$$

Notice that in the previous problem, as usual we have dropped the monotonicity constraint. We will check later that it holds.

From the first order conditions we get that

$$\theta_H (1 - x_H^s) = c \Rightarrow x_H^s = 1 - \frac{c}{\theta_H},$$

thus we have once again that there is no distortion at the top, and the second best quantity for the high type is the same as the first best.

For the low type:

$$\theta_L(1 - x_L^s) - c - \frac{1 - \lambda}{\lambda}(\theta_H(1 - x_L^s) - \theta_L(1 - x_L^s)) = 0$$

which implies that

$$x_L^s = 1 - \frac{c}{\theta_L - \frac{1 - \lambda}{\lambda} \Delta \theta}$$

where $\Delta \theta = \theta_H - \theta_L$. Notice that indeed $x_L^s \leq x_H^s$ and thus the monotonicity constraint is satisfied. The optimal transfers are

$$T_L^s = \theta_L v(x_L^s), \quad T_H^s = \theta_H(v(x_H^s) - v(x_L^s)) + \theta_L v(x_L^s).$$

Let Π_4 be the optimal non-linear tariff.

It is clear from the way we solved the problem, (more restricted to less restricted) that $\Pi_1 \leq \Pi_3 \leq \Pi_4 \leq \Pi_2$.

^a  **Warning!** The following part requires a ridiculous amount of algebra. But for the sake of completeness I will write here the proof that the derivative with respect to λ of the profit difference $\Pi_1 - \Pi'_1$ is strictly positive and thus increasing in λ . Calculating the derivative we get

$$\frac{\partial(\Pi_1 - \Pi'_1)}{\partial \lambda} = \frac{(\theta_H - \theta_L) \left(\frac{\theta_H \theta_L}{\lambda \theta_H - \lambda \theta_L + \theta_L} - c \right)^2}{4 \theta_H \theta_L} + \frac{(\theta_H - \theta_L) \left(\frac{\theta_H \theta_L}{\lambda \theta_H - \lambda \theta_L + \theta_L} - c \right)}{2 (\lambda \theta_H - \lambda \theta_L + \theta_L)}.$$

Expanding all terms, simplifying and canceling terms, then factoring and collecting similar terms yields:

$$\frac{1}{4} \left(c \left(\frac{c}{\theta_L} - 2 \right) - \frac{\theta_H^2 \theta_L}{(\lambda \theta_H - (\lambda - 1) \theta_L)^2} + \theta_H \left(\frac{\theta_L^2}{(\lambda \theta_H - \lambda \theta_L + \theta_L)^2} + 1 \right) \right).$$

We want to show that the previous expression is greater than zero to show that the difference in profits is strictly increasing with respect to λ . The proof will now proceed in two steps: we will show that when $\lambda \rightarrow 0$, the term that we obtain is strictly positive. Next we will show that the derivative is also an increasing function with respect to λ by calculating the second derivative and showing that it is positive. These two facts will allow us to conclude that the first derivative is strictly increasing and thus always strictly positive over its domain.

First note that when $\lambda \rightarrow 0$, the previous expression evaluates to

$$\frac{(c - \theta_H)(c + \theta_H - 2\theta_L)}{\theta_L}.$$

The previous term is strictly positive if and only if $c + \theta_H - 2\theta_L < 0$ which is exactly the condition that the monopoly price with respect to the type- θ_H market is less than the highest willingness to pay of type- θ_H . Now, calculating the second derivative with respect to λ yields:

$$\frac{\theta_H \theta_L (\theta_H - \theta_L)^2}{2 (\lambda \theta_H - \lambda \theta_L + \theta_L)^3},$$

which once again is easy to see is strictly positive. Hence, the first derivative is an increasing function and is strictly positive in all of its domain which concludes the proof.

3. Wuzzle Airlines is the only airline allowed to fly between the Isle of Wuzz and the country of Panda-La. There are two types of passengers, tourists and business. Business travelers are willing to pay more than tourists. The airline, however, cannot tell directly whether a ticket purchaser is a tourist or a business traveler. The two types differ, though, in how much they are willing to pay to avoid having to purchase their tickets in advance. (Passengers do not like to commit themselves in advance to traveling at a particular time.)

More specifically, the utility levels of each of the two types net of the price of the ticket, P , for any given amount of time W prior to the flight that the ticket is purchased are given by

$$\text{Business: } v - \theta_B P - W$$

$$\text{Tourist: } v - \theta_T P - W,$$

where $0 < \theta_B < \theta_T$. (Note that for any given level W , the business traveler is willing to pay more for his ticket. Also, the business traveler is willing to pay more for any given reduction in W .)

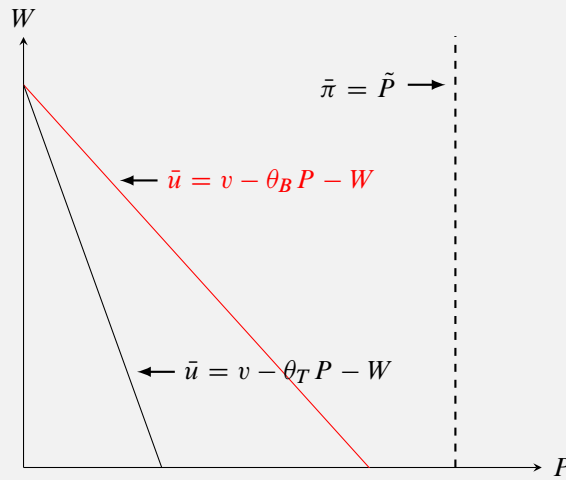
The proportion of travelers who are tourists is λ . Assume that the cost of transporting a passenger is c .

Assume in (a)–(d) that Wuzzle Airlines wants to carry both types of passengers.

- Draw the indifference curves of the two types in (P, W) -space. Draw the airline's isoprofit curves. Now formulate the optimal (profit-maximizing) price discrimination problem mathematically that Wuzzle Airlines would want to solve. *Hint: Impose non negativity of prices as a constraint since, if it charged a negative price, it would sell an infinite number of tickets at this price. Waiting times also need to be nonnegative.*
- Show that in the optimal solution, tourists are indifferent between buying a ticket and not going at all.
- Show that in the optimal solution, business travelers never buy their ticket prior to the flight and are just indifferent between doing this and buying when tourists buy.
- Describe fully the optimal price discrimination scheme under the assumption that they sell to both types. How does it depend on the underlying parameters λ , θ_B , θ_T and c ? Provide an economic interpretation.
- Under what circumstances will Wuzzle Airlines choose to serve only business travelers?

Solution

- The following diagram plots the indifference curves of the two types of travelers and the iso-profit line of the airline in (P, W) -space.



The airline only pays cost c of a passenger if it chooses to provide service, i.e. if it chooses to charge at least one strictly positive price. If by charging at least one strictly positive price the airline finds that it cannot make a profit, then it decides not to operate. Thus, the problem of the airline is

$$\begin{aligned}
 & \max_{\{(P_T, W_T), (P_B, W_B)\}} \max\{\lambda P_T + (1 - \lambda)P_B - c, 0\} & (27) \\
 & \text{subject to} \quad \theta_B P_B + W_B \leq \theta_B P_T + W_T & (\text{IC B}) \\
 & \quad \theta_T P_T + W_T \leq \theta_T P_B + W_B & (\text{IC T}) \\
 & \quad v \geq \theta_B P_B + W_B & (\text{PC B}) \\
 & \quad v \geq \theta_T P_T + W_T & (\text{PC T}) \\
 & \quad P_T \geq 0, P_B \geq 0, W_T \geq 0, W_B \geq 0 & (\text{Non-Negativity})
 \end{aligned}$$

The airline's problem is written in that way because if it decides not to operate, then it doesn't need to pay a cost. The airline doesn't operate when it charges both prices equal to zero. In order to simplify this problem, we will assume for the moment that the airline decides to operate always, i.e. that profit is always positive. Thus we will consider as an objective

$$\lambda P_T + (1 - \lambda)P_B - c,$$

and we will worry later about in what circumstances the firm will make negative profits.

- (b) In the maximization problem above we have that (PC T) and (IC B) together with $\theta_B < \theta_T$ (which in this problem is the single-crossing property) imply that constraint (PC B) is redundant, so it is never binding. constraint (PC T) must therefore be binding for if it would not, then we can reduce P_T and P_B by a small positive quantity $\epsilon > 0$ and all the remaining constraints will still be satisfied and this would improve the airline's profit. The fact that (PC T) is binding implies that tourists are indifferent between buying and not buying a ticket.
- (c) Suppose that $\{(P_T, W_T), (P_B, W_B)\}$ is an optimal, incentive compatible contract. We need to show that $W_B = 0$. Towards a contradiction, assume that $W_B > 0$. Choose $\epsilon > 0$ to reduce W_B by ϵ such that it still remains positive and increase P_B by ϵ/θ_B so that the utility of type B does not change. Now the firm earns higher profits from the business type. The actual change in profit is $(1 - \lambda)\epsilon/\theta_B > 0$. Now we need to check that this new contract we constructed is not attractive for the tourist type. Indeed this is the case:

$$\theta_T P_T + W_T \leq \theta_T P_B + W_B = \theta_T \left(P_B + \frac{\epsilon}{\theta_T} \right) + (W_B - \epsilon) < \theta_T \left(P_B + \frac{\epsilon}{\theta_B} \right) + (W_B - \epsilon),$$

where the first inequality comes from $\{(P_T, W_T), (P_B, W_B)\}$ being an incentive compatible plan, the second equality comes from altering W_B by ϵ but then increasing P_B by ϵ/θ_T so as to not alter the utility of the tourist type, and the third inequality comes from the fact that $\theta_B < \theta_T$. The previous calculations shows that the altered contract is still incentive compatible but it achieves a higher profit for the airline, contradicting the optimality of $\{(P_T, W_T), (P_B, W_B)\}$ which we assumed originally. Thus we must have that $W_B = 0$ at an optimal incentive compatible plan.

Now, if at an optimal contract, business travelers were not indifferent between (P_T, W_T) and (P_B, W_B) , we could slightly raise P_B and all the constraints would remain satisfied, since recall that (PC B) is redundant, and the airline would increase its profit coming from business types. Therefore, in an optimal contract, business types are indifferent between (P_T, W_T) and (P_B, W_B) : this is another way of expressing that (IC B) is *binding*, $\theta_B P_B + W_B = \theta_B P_T + W_T$.

- (d) From part (b) and (c), consider the binding constraints:

$$v = \theta_T P_T + W_T \tag{28}$$

$$\theta_B P_B + W_B = \theta_B P_T + W_T \tag{29}$$

These two constraints together with monotonicity ($P_B \geq P_T$) and single crossing, imply that

$$W_T - W_B = \theta_B P_B - \theta_B P_T = \theta_B (P_B - P_T) \leq \theta_T (P_B - P_T)$$

which is the constraint (PC T).

Thus, once again, we can solve the airline's problem by considering just the binding constraints, the monotonicity constraint, and the non-negativity constraints. From the binding constraints, we will solve for the prices in terms of the remaining waiting time:

$$P_T = \frac{v - W_T}{\theta_T} \tag{30}$$

$$P_B = \frac{v - W_T}{\theta_T} + \frac{W_T}{\theta_B} \tag{31}$$

Note that since $P_T \geq 0$ and $P_B \geq 0$ imply from the previous equations that $W_T \leq v$. Plugging in the previous expressions into the objective of the airline yields the following problem:

$$\max_{W_T} \left\{ \lambda \left(\frac{v - W_T}{\theta_T} \right) + (1 - \lambda) \left(\frac{v - W_T}{\theta_T} + \frac{W_T}{\theta_B} \right) - c \right\} \tag{32}$$

$$\text{subject to } 0 \leq W_T \leq v \tag{33}$$

Note that monotonicity of the prices is automatically satisfied from equations (30) and (31).

We can simplify the objective (32) to get:

$$\frac{v - W_T}{\theta_T} + (1 - \lambda) \frac{W_T}{\theta_B} = \left(\frac{v}{\theta_T} \right) - c = \left(\frac{v}{\theta_T} - c \right) + W_T \left(\frac{1 - \lambda}{\theta_B} - \frac{1}{\theta_T} \right)$$

Remember that we will eventually need to check if it is worth for the firm to operate, but for the moment, the problem of a firm that decides to operate can be reduced to:

$$\max_{W_T \in [0, v]} \left\{ \left(\frac{v}{\theta_T} - c \right) + W_T \left(\frac{1 - \lambda}{\theta_B} - \frac{1}{\theta_T} \right) \right\} \quad (34)$$

If we analyze problem (34) from a purely mathematical perspective, we can notice that the objective function is an *affine function*, i.e. a line with an intercept, and that the maximization is carried out over an interval. Thus, there are only two possible solutions, which depend on the *sign* of the slope $\left(\frac{1 - \lambda}{\theta_B} - \frac{1}{\theta_T} \right)$. The solution is given by:

$$W_T = \begin{cases} v & \text{if } \frac{\theta_T - \theta_B}{\theta_T} \geq \lambda \\ 0 & \text{if } \frac{\theta_T - \theta_B}{\theta_T} < \lambda \end{cases}$$

Before solving for the remaining variables, we need to check under what circumstances the previous is indeed a *true solution* by checking if the firm is making profits. Thus, we will compute first the profits under the two cases above.

$$\text{Profits when } W_T = 0 : \quad \frac{v}{\theta_T} - c$$

$$\text{Profits when } W_T = v : \quad \frac{(1 - \lambda)v}{\theta_B} - c$$

The firm decides to operate whenever the parameters are such that:

$$\min \left\{ \frac{v}{\theta_T}, (1 - \lambda) \frac{v}{\theta_B} \right\} \geq c$$

Whenever the above condition is satisfied, then the remaining variables are given by

$$P_T^* = \begin{cases} 0 & \text{if } \frac{\theta_T - \theta_B}{\theta_T} \geq \lambda \\ \frac{v}{\theta_T} & \text{if } \frac{\theta_T - \theta_B}{\theta_T} < \lambda \end{cases}$$

$$P_B^* = \begin{cases} \frac{v}{\theta_B} & \text{if } \frac{\theta_T - \theta_B}{\theta_T} \geq \lambda \\ \frac{v}{\theta_T} & \text{if } \frac{\theta_T - \theta_B}{\theta_T} < \lambda \end{cases}$$

If the firm decides not to operate, then all the simplifications above (items b–d) do not apply and the optimal contract is $\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \{(0, 0), (0, 0)\}$

Summarizing all these results, the optimal contract is:

$$\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \begin{cases} \left\{ (0, v), \left(\frac{v}{\theta_B}, 0 \right) \right\} & \text{if } \min \left\{ \frac{\theta_T - \theta_B}{\theta_T}, \frac{v - c\theta_B}{v} \right\} \geq \lambda \\ \left\{ \left(\frac{v}{\theta_T}, 0 \right), \left(\frac{v}{\theta_T}, 0 \right) \right\} & \text{if } \lambda > \frac{\theta_T - \theta_B}{\theta_T} \text{ and } \frac{v}{\theta_T} \geq c \\ \{(0, 0), (0, 0)\} & \text{if } \min \left\{ \frac{v}{\theta_T}, (1 - \lambda) \frac{v}{\theta_B} \right\} < c \end{cases} \quad (35)$$

(e) From the optimal solution in (35) we can see that whenever

$$\min \left\{ \frac{\theta_T - \theta_B}{\theta_T}, \frac{v - c\theta_B}{v} \right\} \geq \lambda$$

, the airline is only contracting with the business type:

$$\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \left\{ (0, v), \left(\frac{v}{\theta_B}, 0 \right) \right\}$$

since the price it charges for tourists is zero. If this is the case, we are in a *shutdown contract*.

On the other hand, whenever

$$\lambda > \frac{\theta_T - \theta_B}{\theta_T} \text{ and } \frac{v}{\theta_T} \geq c$$

the airline optimal plan is a *pooling contract*:

$$\{(P_T^*, W_T^*), (P_B^*, W_B^*)\} = \left\{ \left(\frac{v}{\theta_T}, 0 \right), \left(\frac{v}{\theta_T}, 0 \right) \right\}.$$

In the case that the firm can never make positive profits, i.e. when

$$\min \left\{ \frac{v}{\theta_T}, (1 - \lambda) \frac{v}{\theta_B} \right\} < c,$$

then the optimal contract is also a pooling contract in which no traveler gets serviced.